

Worksheet Query Builder

```
1224 -----
1225 -- TASK 5: Package - Employee Management Package
1226 -----
1227 -- Section 9:
1228 -- Employee Management System - Triggers
1229 -- TASK 1: BEFORE INSERT - Auto-Assign emp_ID
1230 CREATE OR REPLACE TRIGGER trg_emp_id
1231 BEFORE INSERT ON employee
1232 FOR EACH ROW
1233 WHEN (NEW.emp_id IS NULL)
1234 BEGIN
1235     :NEW.emp_id := employee_seq.NEXTVAL;
1236 END;
1237 /
1238
1239 -- Test: insert an employee without specifying emp_ID
1240 INSERT INTO employee (fname, lname, gender, age, emp_email, emp_pass, job_id)
1241 VALUES ('Auto', 'Assigned', 'F', 26, 'auto.assigned@ems.com', 'pass123', 2);
1242
1243 SELECT emp_id, fname, lname FROM employee WHERE fname = 'Auto';
1244 -- Expected: emp_id is populated automatically from EMPLOYEE_SEQ
1245
```

Script Output x Query Result x Query Result 1 x

Task completed in 0.053 seconds

PL/SQL procedure successfully completed.

Trigger TRG_EMP_ID compiled

LINE/COL	ERROR
2/3	PL/SQL: Statement ignored
2/16	PLS-00201: identifier 'EMPLOYEE_SEQ.NEXTVAL' must be declared

Errors: check compiler log

Trigger TRG_EMP_ID compiled

1 row inserted.

Compiler - Log

SQL Worksheet History

Worksheet Query Builder

```
1234 BEGIN
1235     :NEW.emp_id := employee_seq.NEXTVAL;
1236 END;
1237 /
1238
1239 -- Test: insert an employee without specifying emp_ID
1240 INSERT INTO employee (fname, lname, gender, age, emp_email, emp_pass, job_id)
1241 VALUES ('Auto', 'Assigned', 'F', 26, 'auto.assigned@ems.com', 'pass123', 2);
1242
1243 SELECT emp_id, fname, lname FROM employee WHERE fname = 'Auto';
1244 -- Expected: emp_id is populated automatically from EMPLOYEE_SEQ
1245 -----
1246 -- TASK 2: AFTER INSERT - Welcome Log
1247 CREATE TABLE employee_log (
1248     log_id          NUMBER          NOT NULL,
1249     emp_id          NUMBER,
1250     action          VARCHAR2(50)    NOT NULL,
1251     log_timestamp   DATE            DEFAULT SYSDATE,
1252     CONSTRAINT pk_employee_log PRIMARY KEY (log_id)
1253 );
1254
1255 CREATE SEQUENCE employee_log_seq START WITH 1 INCREMENT BY 1 NOCACHE;
```

Script Output x Query Result x Query Result 1 x

Task completed in 0.047 seconds

LINE/COL ERROR

2/3 PL/SQL: Statement ignored

2/16 PLS-00201: identifier 'EMPLOYEE_SEQ.NEXTVAL' must be declared

Errors: check compiler log

Trigger TRG_EMP_ID compiled

1 row inserted.

Table EMPLOYEE_LOG created.

Sequence EMPLOYEE_LOG_SEQ created.

Worksheet

Query Builder

1251

log_timestamp DATE

DEFAULT SYSDATE,

1252

CONSTRAINT pk_employee_log PRIMARY KEY (log_id)

1253

);

1254

1255

CREATE SEQUENCE employee_log_seq START WITH 1 INCREMENT BY 1 NOCACHE;

1256

1257

CREATE OR REPLACE TRIGGER trg_emp_welcome_log

1258

AFTER INSERT ON employee

1259

FOR EACH ROW

1260

BEGIN

1261

INSERT INTO employee_log (log_id, emp_id, action, log_timestamp)

1262

VALUES (employee_log_seq.NEXTVAL, :NEW.emp_id, 'NEW HIRE', SYSDATE);

1263

END;

1264

/

1265

-- Test: insert 2 employees and query EMPLOYEE_LOG

1266

INSERT INTO employee (fname, lname, gender, age, emp_email, emp_pass, job_id)

1267

VALUES ('Hire', 'One', 'M', 30, 'hire.one@ems.com', 'pass123', 3);

1268

1269

INSERT INTO employee (fname, lname, gender, age, emp_email, emp_pass, job_id)

1270

VALUES ('Hire', 'Two', 'F', 31, 'hire.two@ems.com', 'pass123', 3);

1271

1272

SELECT * FROM employee_log WHERE action = 'NEW HIRE';

Script Output x

Query Result x

Query Result 1 x

Task completed in 0.031 seconds

1 row inserted.

Table EMPLOYEE_LOG created.

Sequence EMPLOYEE_LOG_SEQ created.

Trigger TRG_EMP_WELCOME_LOG compiled

1 row inserted.

1 row inserted.

Worksheet

Query Builder

1284

DESC employee_log;

1285

1286

CREATE OR REPLACE TRIGGER trg_emp_welcome_log

1287

AFTER INSERT ON employee

1288

FOR EACH ROW

1289

BEGIN

1290

INSERT INTO employee_log (log_id, emp_id, action, log_timestamp)

1291

VALUES (employee_log_seq.NEXTVAL, :NEW.emp_id, 'NEW HIRE', SYSDATE);

1292

END;

1293

/

1294

1295

-- Test: insert 2 employees and query EMPLOYEE_LOG

1296

INSERT INTO employee (fname, lname, gender, age, emp_email, emp_pass, job_id)

1297

VALUES ('Hire', 'One', 'M', 30, 'hire.oraclePer@ems.com', 'pass123', 3);

1298

1299

INSERT INTO employee (fname, lname, gender, age, emp_email, emp_pass, job_id)

1300

VALUES ('Hire', 'Two', 'F', 31, 'hire.oracleOne@ems.com', 'pass123', 3);

1301

1302

SELECT * FROM employee_log WHERE action = 'NEW HIRE';

Script Output x

Query Result x

Query Result 1 x



SQL | All Rows Fetched: 4 in 0.002 seconds

	LOG_ID	EMP_ID	ACTION	LOG_TIMESTAMP
1	1	16	NEW HIRE	17-JUN-26
2	2	17	NEW HIRE	17-JUN-26
3	3	19	NEW HIRE	17-JUN-26
4	4	20	NEW HIRE	17-JUN-26

```
1359
1360 -- Test: delete one employee and verify the archive
1361 -- (using the 'Auto Assigned' test employee from Task 1, after clearing
1362 -- any dependent rows first to satisfy FK constraints)
1363 DELETE FROM payroll WHERE emp_id = (SELECT emp_id FROM employee WHERE fname = 'Auto'
1364 DELETE FROM leave_record WHERE emp_id = (SELECT emp_id FROM employee WHERE fname = '
1365 DELETE FROM qualification WHERE emp_id = (SELECT emp_id FROM employee WHERE fname =
1366 DELETE FROM salary_bonus WHERE emp_id = (SELECT emp_id FROM employee WHERE fname = '
1367 DELETE FROM employee WHERE fname = 'Auto' AND lname = 'Assigned';
1368 COMMIT;
```

Script Output x Query Result x Query Result 1 x

Task completed in 0.06 seconds

0 rows deleted.

0 rows deleted.

1 row deleted.

Commit complete.

```

Worksheet  Query Builder
1326 SELECT salary_id, amount FROM salary_bonds WHERE salary_id = 1,
1327 -----
1328 -- TASK 4: AFTER DELETE - Archive Deleted Employees
1329 CREATE TABLE employee_archive (
1330     emp_id      NUMBER,
1331     fname       VARCHAR2(50),
1332     lname       VARCHAR2(50),
1333     gender      CHAR(1),
1334     age         NUMBER(3),
1335     emp_email    VARCHAR2(100),
1336     emp_pass     VARCHAR2(100),
1337     job_id      NUMBER,
1338     salary_id   NUMBER,
1339     manager_id  NUMBER,
1340     archived_at DATE          DEFAULT SYSDATE,
1341     archived_by VARCHAR2(50)
1342 );
1343
1344 CREATE OR REPLACE TRIGGER trg_archive_employee
1345 AFTER DELETE ON employee
1346 FOR EACH ROW
1347 BEGIN
1348     INSERT INTO employee_archive (
1349         emp_id, fname, lname, gender, age, emp_email, emp_pass,
1350         job_id, salary_id, manager_id, archived_at, archived_by
1351     )
1352     VALUES (
1353         :OLD.emp_id, :OLD.fname, :OLD.lname, :OLD.gender, :OLD.age,
1354         :OLD.emp_email, :OLD.emp_pass, :OLD.job_id, :OLD.salary_id,
1355         :OLD.manager_id, SYSDATE, USER
1356     );
1357 END;
1358 /

```

Script Output x Query Result x Query Result 1 x

Task completed in 0.052 seconds

Table EMPLOYEE_ARCHIVE created.

Trigger TRG_ARCHIVE_EMPLOYEE compiled

```

1369
1370 SELECT * FROM employee_archive;

```

Script Output x Query Result x Query Result 1 x

All Rows Fetched: 1 in 0.021 seconds

EMP_ID	FNAME	LNAME	GENDER	AGE	EMP_EMAIL	EMP_PASS	JOB_ID	SALARY_ID	MANAGER_ID	ARCHIVED_AT	ARCHIVED_BY
1	15 Auto	Assigned	F	26	auto.assigned@ems.com	pass123	2	(null)	(null)	17-JUN-26	EMS